

# DW\_sqrt\_pipe

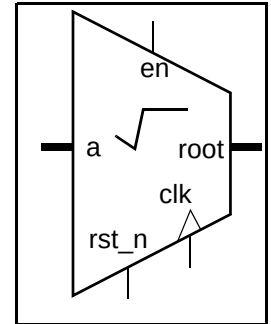
## Stallable Pipelined Square Root

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

### Features and Benefits

- Re-uses the DW\_sqrt component and adds pipeline structures
- Parameterized word length
- Unsigned and signed (two's complement) data operation
- Parameterized number of pipeline stages
- Parameterized stall mode (stallable or non-stallable)
- Parameterized reset mode (no reset, asynchronous or synchronous reset)
- Automatic pipeline retiming
- Provides minPower benefits (for details, see [Table 1-3](#) on page 2)

### Revision History



### Description

DW\_sqrt\_pipe is a universal stallable pipelined square root generator. DW\_sqrt\_pipe computes the square root of operand *a* with a latency of *num\_stages* – 1 clock cycles.

Table 1-1 Pin Description

| Pin Name | Width                      | Direction | Function                                                                                                                                                                 |
|----------|----------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| clk      | 1 bit                      | Input     | Input clock                                                                                                                                                              |
| rst_n    | 1 bit                      | Input     | Reset, active-low (not used if parameter <i>rst_mode</i> = 0)                                                                                                            |
| en       | 1 bit                      | Input     | Register enable, active high (used only if parameter <i>stall_mode</i> = 1) <ul style="list-style-type: none"> <li>■ 0 = Stall</li> <li>■ 1 = Enable register</li> </ul> |
| a        | <i>width</i> bits          | Input     | Radicand                                                                                                                                                                 |
| root     | ( <i>width</i> + 1)/2 bits | Output    | Square root                                                                                                                                                              |

Table 1-2 Parameter Description

| Parameter | Values               | Description             |
|-----------|----------------------|-------------------------|
| width     | ≥ 2<br>Default: None | Word length of <i>a</i> |

**Table 1-2 Parameter Description (Continued)**

| Parameter   | Values               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| num_stages  | ≥ 2<br>Default: None | Number of pipeline stages                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| stall_mode  | 0 or 1<br>Default: 1 | Stall mode <ul style="list-style-type: none"> <li>0: Non-stallable</li> <li>1: Stallable</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                    |
| rst_mode    | 0 to 2<br>Default: 1 | Reset mode <ul style="list-style-type: none"> <li>0: No reset</li> <li>1: Asynchronous reset</li> <li>2: Synchronous reset)</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| tc_mode     | 0 or 1<br>Default: 0 | Two's complement mode <ul style="list-style-type: none"> <li>0: Unsigned number</li> <li>1: Signed number</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |
| op_iso_mode | 0 to 4<br>Default: 0 | Operand isolation mode (controls datapath gating for minPower flow)<br>Allows you to set the style of minPower datapath gating for this module <ul style="list-style-type: none"> <li>0: Use the DW_lp_op_iso_mode<sup>a</sup> synthesis setting</li> <li>1: 'none'</li> <li>2: 'and'</li> <li>3: 'or'</li> <li>4: Preferred gating style: 'and'</li> </ul> For details about enabling minPower datapath gating for this component, see <a href="#">“Enabling minPower”</a> on page 6. |

- a. The DW\_lp\_op\_iso\_mode synthesis variable is available only in Design Compiler.  
DW\_lp\_op\_iso\_mode sets a global style of datapath gating. To use the global style, set *op\_iso\_mode* to '0'. Note that if the *op\_iso\_mode* parameter is set to '0' and DW\_lp\_op\_iso\_mode is either not set or set to 0', then no datapath gating is inserted for this component.

**Table 1-3 Synthesis Implementations**

| Implementation Name | Implementation                | License Feature Required                                                                                                     |
|---------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| str                 | Pipelined str synthesis model | DesignWare                                                                                                                   |
| lpwr <sup>a</sup>   | Low Power synthesis model     | <ul style="list-style-type: none"> <li>DesignWare (P-2019.03 and later)</li> <li>DesignWare-LP (before P-2019.03)</li> </ul> |

- a. Requires that you enable minPower; for details, see [“Enabling minPower”](#) on page 6.  
When minPower is enabled, the lpwr implementation is always chosen during synthesis.

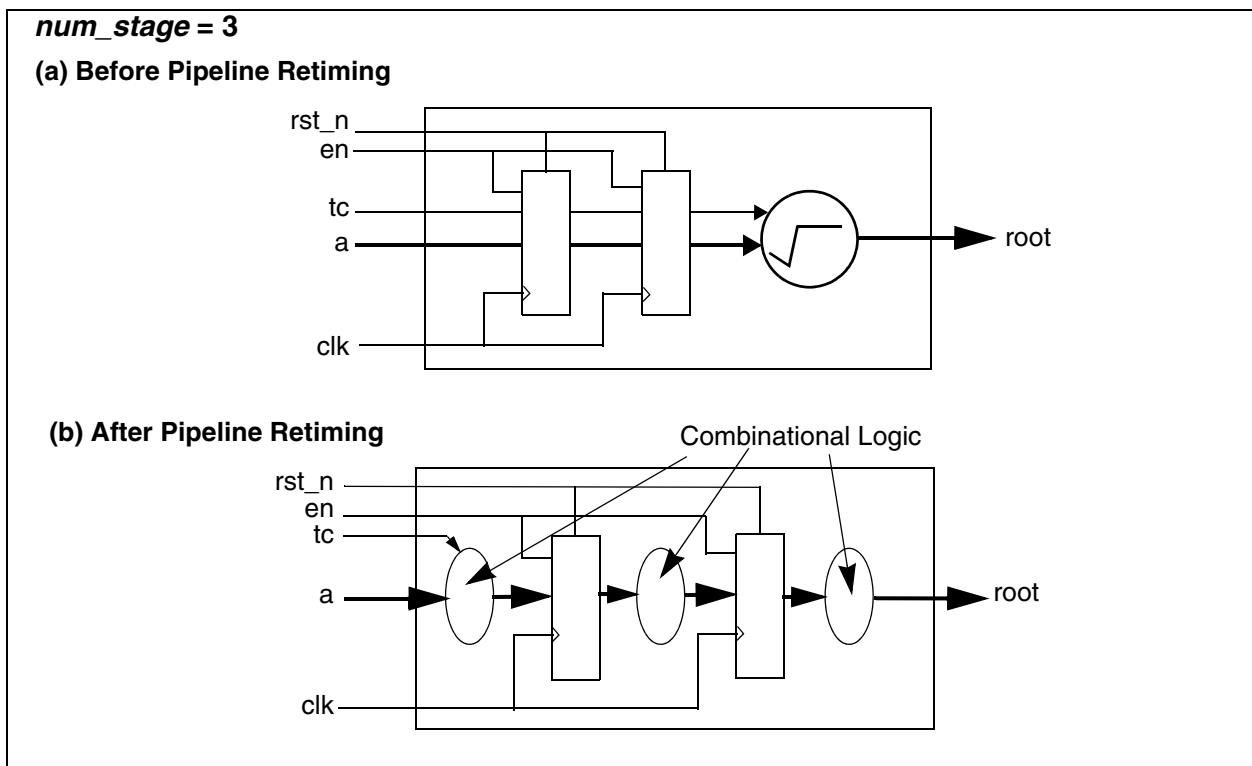
**Table 1-4 Simulation Models**

| Model                            | Function                             |
|----------------------------------|--------------------------------------|
| DW02.DW_SQRT_PIPE_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw02/src/DW_sqrt_pipe_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW_sqrt_pipe.v        | Verilog simulation model source code |

## Functional Description

The parameter *tc\_mode* determines whether the input and output data is interpreted as unsigned (*tc\_mode* = 0) or signed (*tc\_mode* = 1) numbers.

Automatic pipeline retiming ensures optimal placement of pipeline registers within the square root generator to achieve maximum throughput. The pipeline can be stalled by setting the load enable signal *en* = 0 (when *stall\_mode* = 1). The pipeline registers can either have no reset (*rst\_mode* = 0) or an asynchronous (*rst\_mode* = 1) or synchronous reset (*rst\_mode* = 2) connected to the reset signal *rst\_n*.

**Figure 1-1 Pipeline Retiming**

## Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW\_SUPPRESS\_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

+define+DW\_SUPPRESS\_WARN (which is used for the Synopsys VCS simulator)

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW\_DISABLE\_CLK\_MONITOR macro. You can define this macro in the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_DISABLE_CLK_MONITOR
```

- Or, include a command line option to the simulator, such as:

+define+DW\_DISABLE\_CLK\_MONITOR (which is used for the Synopsys VCS simulator)

This message is also suppressed using the DW\_SUPPRESS\_WARN macro explained earlier.

Timing Waveform

Figure 1-2 Waveform 1

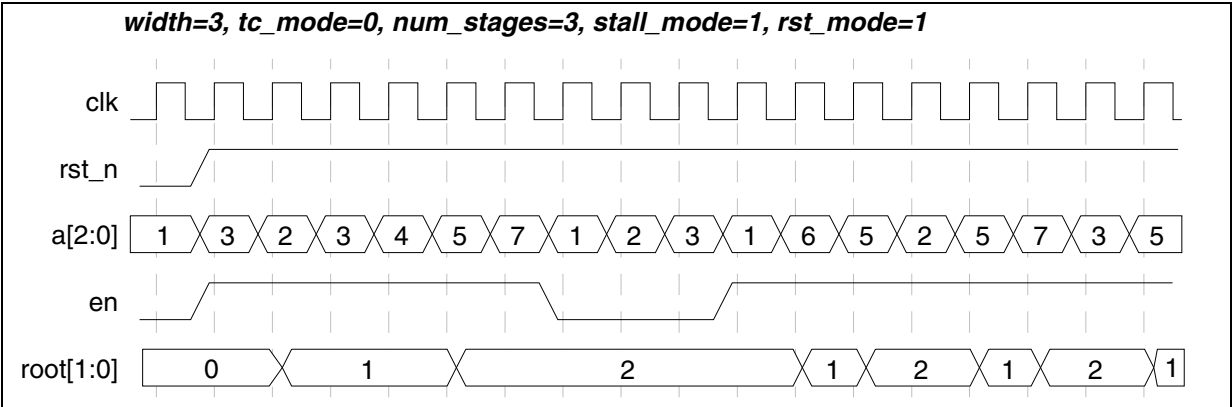
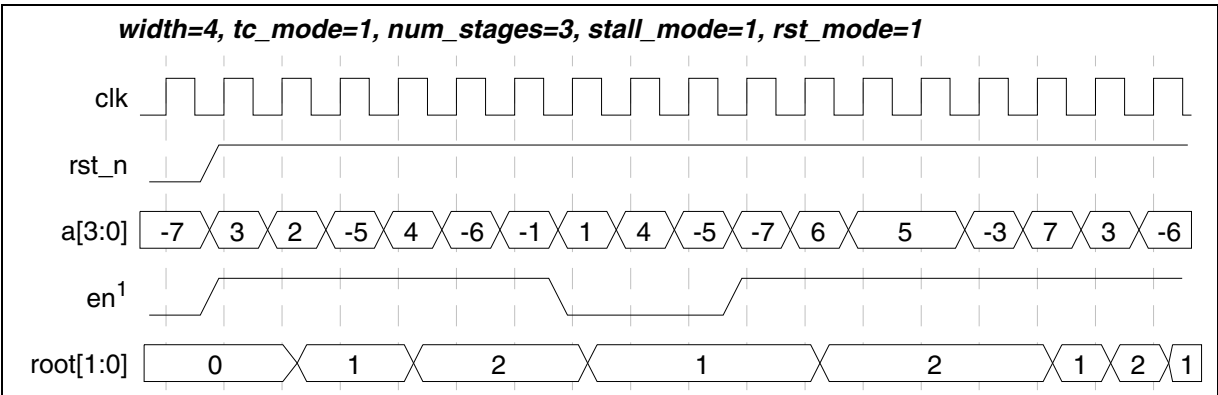


Figure 1-3 Waveform 2



<sup>1</sup>If parameter stall\_mode=0, then pin en has no effect.

Figure 1-4 Waveform 3

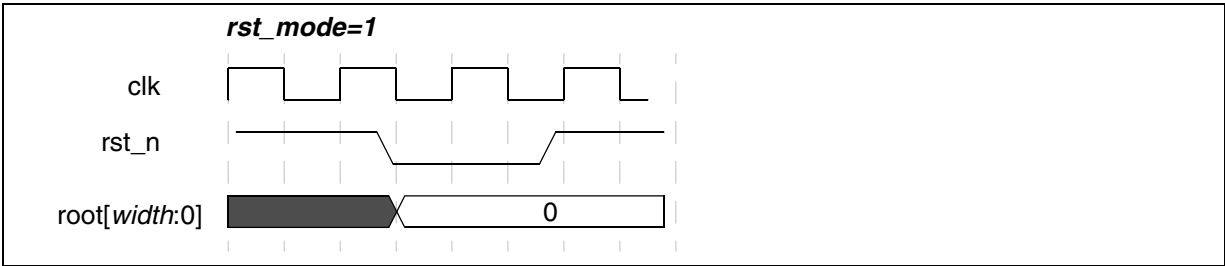
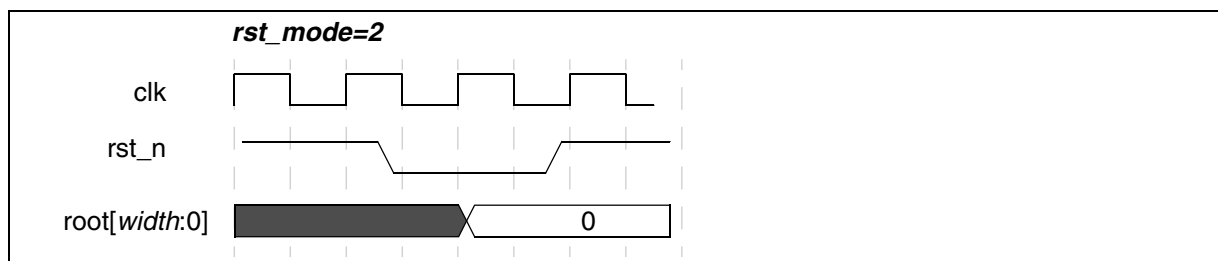


Figure 1-5 Waveform 4



## Enabling minPower

You can instantiate this component without enabling minPower, but to achieve datapath gating power savings, you must enable minPower optimization, as follows:

- Design Compiler

- Version P-2019.03 and later:

```
set power_enable_minpower true
```

- Before version P-2019.03 (requires the DesignWare-LP license feature):

```
set synthetic_library {dw_foundation.sldb dw_minpower.sldb}
set link_library {* $target_library $synthetic_library}
```

- Fusion Compiler

Optimization for minPower is enabled as part of the total\_power metric setting. To enable the total\_power metric, use the following:

```
set_qor_strategy -stage synthesis -metric total_power
```

## Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```

library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_sqrt_pipe_inst is
  generic (inst_width      : POSITIVE := 2; inst_tc_mode      : NATURAL := 0;
           inst_num_stages : POSITIVE := 2; inst_stall_mode   : NATURAL := 1;
           inst_rst_mode   : NATURAL := 1; inst_op_iso_mode   : NATURAL := 0 );
  port (inst_clk   : in std_logic;
        inst_rst_n : in std_logic;
        inst_en    : in std_logic;
        inst_a     : in std_logic_vector(inst_width-1 downto 0);
        root_inst  : out std_logic_vector((inst_width+1)/2-1 downto 0) );
end DW_sqrt_pipe_inst;

architecture inst of DW_sqrt_pipe_inst is
begin
  -- Instance of DW_sqrt_pipe
  U1 : DW_sqrt_pipe
    generic map (width => inst_width, tc_mode => inst_tc_mode,
                 num_stages => inst_num_stages, stall_mode => inst_stall_mode,
                 rst_mode => inst_rst_mode, op_iso_mode => inst_op_iso_mode )
    port map (clk => inst_clk, rst_n => inst_rst_n, en => inst_en,
              a => inst_a, root => root_inst );
end inst;

-- Configuration for use with VSS simulator
-- pragma translate_off
configuration DW_sqrt_pipe_inst_cfg_inst of DW_sqrt_pipe_inst is
  for inst
  end for; -- inst
end DW_sqrt_pipe_inst_cfg_inst;
-- pragma translate_on

```

## HDL Usage Through Component Instantiation - Verilog

```
module DW_sqrt_pipe_inst( inst_clk, inst_rst_n, inst_en, inst_a, root_inst );
    parameter inst_width = 2;
    parameter inst_tc_mode = 0;
    parameter inst_num_stages = 2;
    parameter inst_stall_mode = 1;
    parameter inst_rst_mode = 1;
    parameter inst_op_iso_mode = 0;

    input inst_clk;
    input inst_rst_n;
    input inst_en;
    input [inst_width-1 : 0] inst_a;
    output [(inst_width+1)/2-1 : 0] root_inst;

    // Instance of DW_sqrt_pipe
    DW_sqrt_pipe #(inst_width, inst_tc_mode, inst_num_stages,
                  inst_stall_mode, inst_rst_mode, inst_op_iso_mode)
        U1 (.clk(inst_clk), .rst_n(inst_rst_n),
           .en(inst_en), .a(inst_a), .root(root_inst) );
endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                                                                                                                                       |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | ■ Updated date and version; no technical changes                                                                                                                                                                                                                                              |
| December 2023  | DWBB_202312.0 | ■ Updated date and version; no technical changes                                                                                                                                                                                                                                              |
| July 2020      | DWBB_201912.5 | ■ Adjusted content and title of “ <a href="#">Suppressing Warning Messages During Verilog Simulation</a> ” on page 4 and added the DW_SUPPRESS_WARN macro                                                                                                                                     |
| October 2019   | DWBB_201903.5 | ■ Added the “Disabling Clock Monitor Messages” section                                                                                                                                                                                                                                        |
| March 2019     | DWBB_201903.0 | ■ Clarified the op_iso_mode parameter in <a href="#">Table 1-2</a> on page 1<br>■ Clarified license requirements in <a href="#">Table 1-3</a> on page 2<br>■ Added “ <a href="#">Enabling minPower</a> ” on page 6<br>■ Added this Revision History table and the document links on this page |

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